

X is a continuous random variable if there exists a nonnegative function f , defined for all reals $x \in (-\infty, \infty)$ having the property that for any set B of real numbers,

$$P(X \in B) = \int_B f(x) dx, \quad f \text{ is known as a probability density function (pdf) of } X$$

We have $1 = P(X \in (-\infty, \infty)) = \int_{-\infty}^{\infty} f(x) dx$
 Letting $B = [a, b]$, we obtain

$$P\{a \leq X \leq b\} = \int_a^b f(x) dx$$

Letting $a=b$, we get

$$P\{X=a\} = \int_a^a f(x) dx = 0, \quad \text{for any fixed value, Probability of that is 0}$$

For continuous variables, the cumulative distribution function (cdf) is calculated by integrating the pdf from negative infinity up to a .

$$F(a) = P\{X \in (-\infty, a]\} = \int_{-\infty}^a f(x) dx$$

Differentiating both sides yields

$$\frac{d}{da} F(a) = f(a), \quad \text{the density is the derivative of the cdf}$$

ex If X is continuous with distribution function F_x and density function f_x , find the density function $Y=2X$.

$$\begin{aligned} F_Y(a) &= P\{Y \leq a\} \\ &= P\{2X \leq a\} \\ &= P\{X \leq a/2\} \\ &= F_X(a/2) \end{aligned}$$

Differentiation gives

$$f_Y(a) = \frac{1}{2} f_X(a/2)$$

5.2 Expectation and Variance of Continuous Random Variables

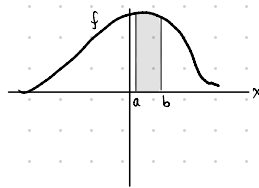
For discrete r.v., $E[X] = \sum_x x P(X=x)$

For continuous r.v with density function $f(x)$, $E[X] = \int_{-\infty}^{\infty} x f(x) dx$

ex Find $E[X]$ of X when pdf f is

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow \int_0^1 x f(x) dx = \int_0^1 2x^2 dx = \frac{2}{3}$$



exp Given pdf

$$f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{else} \end{cases}, \text{ find } E[e^X]$$

recall that

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

Let $Y = e^X$. $F_Y = P(Y \leq x)$ (F_Y is the cdf of Y)

$$\begin{aligned}
 &= P(e^X \leq x) \\
 &= P(X \leq \log x) \\
 &= \int_0^{\log x} f_Y(y) dy \\
 &= \log(x)
 \end{aligned}$$

$$f_Y(x) = \frac{1}{x} \quad 1 \leq x \leq e \quad \text{so} \quad E[e^X] = E[Y] = \int_{-\infty}^{\infty} x f_Y(x) dx = \int_1^e dx = e - 1$$

Note the cdf method used:

1. write cdf definition of Y : $F_Y(y) = P(Y \leq y)$
2. substitute $g(X)$ for Y , isolate X : $P(g(X) \leq y) = P(X \leq g^{-1}(y))$
3. express in terms of X 's distribution and differentiate: $F_Y(y) = \int_{-\infty}^{g^{-1}(y)} f_X(x) dx$

Proposition: (Law of Unconscious Statistician)

$$\hookrightarrow \frac{d}{dy} \left[\int_{-\infty}^{g^{-1}(y)} f_X(x) dx \right]$$

If X is continuous r.v. with pdf $f(x)$,
then

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

As seen above, $E[e^X] = \int_0^1 e^x dx = e - 1$

Lemma: (Tail Sum Formula) For any nonneg. r.v. Y , $E[Y] = \int_0^{\infty} P(Y > y) dy$

Proof: Let Y have pdf f_Y . We have

$$\begin{aligned}
 \int_0^{\infty} P(Y > y) dy &= \int_0^{\infty} \int_y^{\infty} f_Y(x) dx dy \\
 &= \int_0^{\infty} \int_0^x dy f_Y(x) dx \quad (\text{Fubini's Theorem}) \\
 &= E[Y]
 \end{aligned}$$

spread out evenly

exp A stick of length 1 is split at a point U with density function $f_U(u) = 1$, $0 < u < 1$
Determine expected length of piece that contains the point p , $0 \leq p \leq 1$

Let $L_p(U)$ denote length of the substick that contains point p . Note $L_p(U) = \begin{cases} 1-u, & u < p \\ u, & u > p \end{cases}$

From proposition above,

$$\begin{aligned}
 E[L_p(U)] &= \int_0^1 L_p(u) du \\
 &= \int_0^p (1-u) du + \int_p^1 u du \\
 &= \frac{1}{2} - \frac{(1-p)^2}{2} + \frac{1}{2} - \frac{p^2}{2} \\
 &= \frac{1}{2} + p(1-p)
 \end{aligned}$$

Corollary: If a and b are constants, then

$$E[aX+b] = aE[X] + b \quad (\text{same as discrete})$$

If X is r.v. with $E[X] = \mu$,

$$\text{Var}(X) = E[(X-\mu)^2] \quad \text{or} \quad \text{Var}(X) = E[X^2] - E[X]^2$$

exp Find $\text{Var}(X)$ in $f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$

recall $E[X] = \frac{2}{3}$ · $E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx$

$$\frac{1}{2} - \left(\frac{2}{3}\right)^2 = \boxed{\frac{1}{18}}$$

$$= \int_0^1 2x^3 dx$$

$$= \frac{1}{2}$$

↳ Note Law of Unconscious Statistician (LOTUS)

Also $\text{Var}(aX+b) = a^2 \text{Var}(X)$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

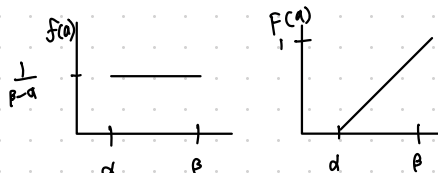
5.3 The Uniform r.v.

A r.v. is said to be uniformly distributed over $(0,1)$ if its pdf is given by

$$f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{else} \end{cases}$$

In general, X is uniform r.v. on interval (α, β) if pdf of X is given by

$$f(x) = \begin{cases} \frac{1}{\beta-\alpha} & \text{if } \alpha < x < \beta \\ 0 & \text{otherwise} \end{cases}$$



Since $F(a) = \int_{-\infty}^a f(x) dx$, the cdf of a function of uniform r.v. is

$$F(a) = \begin{cases} 0 & a \leq \alpha \\ \frac{a-\alpha}{\beta-\alpha} & \alpha < a < \beta \\ 1 & a \geq \beta \end{cases}$$

Let X be uniform r.v. over (α, β) . Find $E[X]$ & $\text{Var}(X)$.

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_{\alpha}^{\beta} \frac{x}{\beta-\alpha} dx \Rightarrow \frac{\beta+\alpha}{2}$$

$$\text{Var}(X) = E[X^2] + E[X]^2$$

$$E[X^2] = \int_a^b x^2 f(x) dx = \int_a^b \frac{x^2}{\beta - \alpha} dx \Rightarrow \frac{\beta^2 + \alpha\beta + \alpha^2}{3}$$

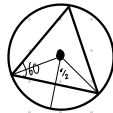
$$\text{Var}(X) = \frac{\beta^2 + \alpha\beta + \alpha^2}{3} - \frac{(\alpha + \beta)^2}{4} = \frac{(\beta - \alpha)^2}{12}$$

Bertrand's paradox

exp Consider a chord on a circle. What is the probability that the length of the chord will be greater than an inscribed equilateral triangle?

The position of the chord is determined by its perpendicular distance D to the center.

$$P(D < \frac{r}{2}) = \frac{r/2}{r} = \frac{1}{2}$$



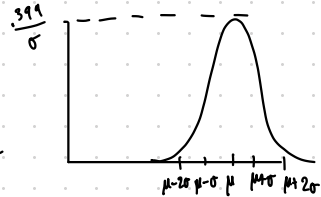
recall 30-60-90 triangle properties

5.4 Normal Random Variables

X is a normal r.v. (X is normally distributed) with parameters μ and σ^2 if the density of X is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

If X is normally distributed, then $Y = aX + b$ is normally distributed with parameters $a\mu + b$ and $a^2\sigma^2$



(a linear transformation of X is closed)

If X is normally distributed with parameters μ and σ^2 , then $Z = \frac{X - \mu}{\sigma}$ is normally distributed with parameters 0 and 1.

↳ Known as a unit, or a standard

Any general normal cdf can be evaluated by standardizing x :

$$F(x) = \Phi\left(\frac{x - \mu}{\sigma}\right) \quad \text{where } \Phi \text{ is the standard normal dist } Z \sim N(0, 1)$$

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt \quad \text{cdf of a standard normal r.v. } \Phi$$

Finding $E[X]$.

$$\text{By def, } E[X] = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$\text{Let } z = \frac{x - \mu}{\sigma}. \quad x = \sigma z + \mu \quad \text{and} \quad dx = \sigma dz$$

$$E[X] = \int_{-\infty}^{\infty} (\sigma z + \mu) \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

using Z helps w
shift center from μ to 0
and variance from σ^2 to 1
or

$$X \sim N(\mu, \sigma^2) \Rightarrow Z \sim N(0, 1)$$

$$= \underbrace{\sigma \int_{-\infty}^{\infty} \frac{z}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{\text{odd function} = 0} + \underbrace{\mu \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{\text{standard normal dist}}$$

Thus,

$$E[X] = \mu$$

$$\text{Var}(X) = E[(X-\mu)^2]$$

$$= \int_{-\infty}^{\infty} (x-\mu)^2 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Let $z = \frac{x-\mu}{\sigma}$ again and we get

$$= \int_{-\infty}^{\infty} \sigma^2 z^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = \frac{\sigma^2}{2\pi} \int_{-\infty}^{\infty} z^2 e^{-\frac{z^2}{2}} dz$$

Integrate by Parts and we end with $\boxed{\text{Var}(X) = \sigma^2}$

Since $Z = \frac{X-\mu}{\sigma}$ is a standard normal r.v. whenever X is normally distrib. with params μ and σ^2 , it follows that the distribution function of X is

$$F_X(a) = P(X \leq a) = P\left(\frac{X-\mu}{\sigma} \leq \frac{a-\mu}{\sigma}\right) = \Phi\left(\frac{a-\mu}{\sigma}\right)$$

exp An exam is good if it is normally dist. The instructor uses the scores to find μ and σ^2 and gives A to those who score above $\mu+\sigma$, B to those between μ and $\mu+\sigma$, C in between $\mu-\sigma$ to μ , D $\mu-2\sigma$ to $\mu-\sigma$, and F under $\mu-2\sigma$.

$$\text{Since } P(X > \mu + \sigma) = P\left(\frac{X-\mu}{\sigma} > 1\right) = 1 - \Phi(1) \approx .16 \text{ or } 16\% \quad A$$

$$P(\mu < X < \mu + \sigma) = P\left(0 < \frac{X-\mu}{\sigma} < 1\right) = \Phi(1) - \Phi(0) \approx .34 \quad 34\% \quad B$$

$$\text{By the nature of } N, \quad P(\mu - \sigma < X < \mu) \approx .34 \quad 34\% \quad C$$

$$P(\mu - 2\sigma < X < \mu - \sigma) = P(-2 < \frac{X-\mu}{\sigma} < -1) = \Phi(-1) - \Phi(-2) \approx .14 \quad 14\% \quad D$$

$$P(X < \mu - 2\sigma) = P\left(\frac{X-\mu}{\sigma} < -2\right) = \Phi(-2) \approx .02 \quad 2\% \quad F$$

exp $\mu = 270, \sigma^2 = 100$

$$\begin{aligned} P(X > 290 \text{ or } X < 240) &= P(X > 290) + P(X < 240) \\ &= P\left(\frac{X-270}{10} > 2\right) + P\left(\frac{X-270}{10} < -3\right) \\ &= 1 - \Phi(2) + 1 - \Phi(3) \\ &\approx .0241 \end{aligned}$$

stand andize

exp In finance, the VAR (value at risk) is defined as a value v s.t. there is a 1% chance that the loss from investment is greater than v . If the gain from investment X is $N(\mu, \sigma^2)$ then because VAR is loss is equal to negative of that gain, the VAR of such an investment is equal to

$$.01 = P(-X > v)$$

$$.01 = P\left(\frac{-X + \mu}{\sigma} > \frac{v + \mu}{\sigma}\right) = 1 - \Phi\left(\frac{v + \mu}{\sigma}\right). \quad \text{Since } \Phi(2.33) = .99,$$

$$\frac{v + \mu}{\sigma} = 2.33, \text{ that is, } v = 2.33\sigma - \mu$$

5.4.1 The Normal Approximation to the binomial distribution

DeMoivre-Laplace limit theorem states that when n is large, a binomial r.v. with parameters n and p will have approximately the same distribution as a normal r.v. with same mean and variance

Formally, if S_n denotes the # of successes that occur in n independent trials, each success a probability p , then for any $a < b$,

$$P\left\{a \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq b\right\} \rightarrow \Phi(b) - \Phi(a)$$

↳ note that np is $E(X)$, or μ , and $\sqrt{np(1-p)}$ is σ or std.

(Now we have 2 approx. to the binomial dist, the Poisson and normal)

exp X is number of heads in 40 flips. $P(X=20)$?

For binomial: $P(X=20) = \binom{40}{20} \left(\frac{1}{2}\right)^{40} \approx .1254$

Normal: $P(X=20) = P(19.5 < X < 20.5)$
 $= P\left(\frac{19.5-20}{\sqrt{10}} < \frac{X-20}{\sqrt{10}} < \frac{20.5-20}{\sqrt{10}}\right)$

$$= P(-.16 < \frac{X-20}{\sqrt{10}} < .16)$$

$$= \Phi(.16) - \Phi(-.16) \approx .1272$$

• Continuity correction: to fix mismatch between discrete & continuous.

$X=k$ is adjusted by $k \pm 0.5$

exp Ideal class size is 150. Back then, only 30% accepted actually attend. So they accept 450 students. Probability that over 150 accept this year?

Let X denote number of students that attend. Then $X \sim \text{Bin}(450, .3)$

$$P(X \geq 150.5) = P\left(\frac{X - (450)(.3)}{\sqrt{450 \cdot .3 \cdot .7}} \geq \frac{150.5 - 450 \cdot .3}{\sqrt{450 \cdot .3 \cdot .7}}\right)$$

$$\approx 1 - \Phi(1.59)$$

$$\approx .0559$$

$\Rightarrow < 6\%$ chance

5.5 Exponential Random Variables

A continuous r.v whose pdf is given for some $\lambda > 0$ by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad \text{is an exponential r.v with parameter } \lambda$$

The cdf is given by $F(a) = P(X \leq a)$

$$= \int_0^a \lambda e^{-\lambda x} dx$$

$$= -e^{-\lambda x} \Big|_0^a = 1 - e^{-\lambda a}$$

▲ Note tail probability:

$$P(X > a) = 1 - F(a) = e^{-\lambda a}$$

$$- E[X^n] = \int_0^\infty x^n \lambda e^{-\lambda x} dx, \quad \text{integrate by parts yields}$$

$$\Rightarrow \frac{n}{\lambda} E[X^{n-1}], \quad \text{so } E[X] = \frac{1}{\lambda} \text{ and } E[X^2] = \frac{2}{\lambda} E[X] = \frac{2}{\lambda^2}$$

$$\text{Hence, } \text{Var}(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Conceptually this makes sense. Say the rate parameter (λ) is 2 customers per minute. The average wait time would be $\frac{1}{\lambda} = \frac{1}{2}$ minutes or .5 minutes

In practice, the exponential distribution arises as a distribution of the amount of time until some specific event occurs.

- earthquake, or war breaks out, or until a call you receive is from wrong number

← Exponential functions are memoryless: the remaining wait time until an event occurs does not depend on the time elapsed.

Mathematically, X is exponentially distributed, for any $s, t \geq 0$

$$P(X > s+t \mid X > s) = P(X > t)$$

Concretely, imagine you work at a coffee shop. Customers arrive follow an exponential distribution with an average of 1 customer every 10 minutes. ($\lambda = .1$ per min) You open at 8:00 AM, the probability of waiting at least 15 minutes ($> 8:15$) for first customer is

$$P(X > 15) = \int_{15}^{\infty} .1 e^{-.1t} dt = e^{-.1 \cdot 15} \approx 22.3\%$$

It's 8:20 now and no customer arrived. What's the probability a customer walks in now?

$$\Rightarrow P(X > 35 \mid X > 20) = P(X > 15) \approx 22.3\%$$

exp Suppose a post office has 2 clerks. When person A enters, both clerks are busy. Person A is next in line. If the amount of time that a clerk spends with someone is exponentially distributed with parameter λ . Probability that person A is the last to leave.

Consider the time when person A gets to a clerk. At this point, the other clerk is still in service. Because exponential is memoryless, it follows that the additional amount of time that this other person would still have to spend is exponentially distributed with λ . That is, it's as if he just started his service; hence person A finishing before any of these 2 is 50%.

exp Suppose # of miles a car battery can run before dying is around 10k miles. If a person desires to take a 5000 mile trip, what is the probability that he will complete the trip without replacing the car battery?

$$\text{Since } E[X] = 10k, \lambda = \frac{1}{E[X]} = \frac{1}{10000} \text{ miles}^{-1}$$

$$P(X > 5000) = e^{-\lambda \times 5000} = e^{-.5} \approx 60\%$$

5.5.1 Hazard Rate Function

Continuous r.v. X that is interpreted as the lifetime of some item. Let X have distribution function F and density f . The hazard rate - or failure rate - is defined by

$$\lambda(t) = \frac{f(t)}{\bar{F}(t)} \quad \bar{F} = 1 - F$$

$$\text{Consider } P(X \in (t, t+dt) | X > t) = \frac{P(X \in (t, t+dt), X > t)}{P(X > t)}$$

$$= \frac{P(X \in (t, t+dt))}{P(X > t)} \approx \frac{f(t)}{\bar{F}(t)} dt$$

Note that $\lambda(t) = \lambda$ in an exponential distribution (it's constant)

The failure rate uniquely determines the cdf for a continuous non-neg r.v.

$$F(t) = 1 - e^{-\int_0^t \lambda(s) ds}$$

$$\text{For instance, if } \lambda(t) = a + bt \text{ then } F(t) = 1 - e^{-at - bt^2/2}$$

$$\text{Consequently, we get the pdf: } f(t) = (a + bt) e^{-(at - bt^2/2)} \quad t \geq 0$$

exp One hears that the death rate of smokers is 2x that of a nonsmoker. What does this mean?

If $\lambda_s(t)$ denotes hazard rate of smoker at age t and $\lambda_n(t)$ is nonsmoker, then the statement is equivalent to

$$\lambda_s(t) = 2\lambda_n(t)$$

The probability that an A -year-old nonsmoker will survive until B is

$$P(A\text{-year-old nonsmoker reaches age } B) =$$

$$= P(\text{nonsmoker's lifetime} > B \mid \text{nonsmoker's lifetime} > A)$$

$$= \frac{1 - F_{\text{non}}(B)}{1 - F_{\text{non}}(A)} = \frac{\exp \left\{ -\int_0^B \lambda_n(t) dt \right\}}{\exp \left\{ -\int_0^A \lambda_n(t) dt \right\}} = \exp \left\{ -\int_A^B \lambda_n(t) dt \right\}$$

$$\begin{aligned} P(\text{A old smoker reaches B}) &= \exp \left\{ -\int_A^B \lambda_s(t) dt \right\} \\ &= \exp \left\{ -2 \int_A^B \lambda_n(t) dt \right\} \\ &= \left[\exp \left\{ -\int_A^B \lambda_n(t) dt \right\} \right]^2 \end{aligned}$$

So, for 2 people of same age, the probability a smoker survives to any given age is square root that of the non smoker.

For instance, if $\lambda_n(t) = \frac{1}{30}$, $50 \leq t \leq 60$, then nonsmoker is $e^{-\frac{1}{3}} \approx .7165$
 Smoker is $e^{-\frac{2}{3}} \approx .5134$

5.6 Other Continuous Distributions

• The Gamma Distribution

- parameters (α, λ)
- density function

$$f(x) = \begin{cases} \frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

where $\Gamma(\alpha)$ is a gamma function, defined as

$$\Gamma(\alpha) = \int_0^{\infty} e^{-y} y^{\alpha-1} dy$$

Integrating gives us.

$$\begin{aligned} \Gamma(\alpha) &= -e^{-y} y^{\alpha-1} \Big|_0^{\infty} + \int_0^{\infty} e^{-y} (\alpha-1) y^{\alpha-2} dy \quad (\text{by parts}) \\ &= (\alpha-1) \int_0^{\infty} e^{-y} y^{\alpha-2} dy \\ &= (\alpha-1) \Gamma(\alpha-1) \end{aligned}$$

So, for a value $\alpha = n$, we obtain

$$\begin{aligned} \Gamma(n) &= (n-1) \Gamma(n-1) \\ &= (n-1)(n-2) \Gamma(n-2) \\ &= \dots \\ &= (n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot \Gamma(1) \end{aligned}$$

Since $\Gamma(1) = \int_0^{\infty} e^{-x} dx = 1$, it follows that

$$\Gamma(n) = (n-1)!$$

The gamma distribution often arises as the distribution of the amount of time one has to wait until a total of n events occurred

$$\begin{aligned} E[X] &= \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \lambda x e^{-\lambda x} (\lambda x)^{\alpha-1} dx \\ &= \frac{1}{\lambda \Gamma(\alpha)} \int_0^{\infty} \lambda e^{-\lambda x} (\lambda x)^{\alpha} dx \quad \left\{ \begin{array}{l} x = \frac{1}{\lambda} \lambda x \\ \lambda x e^{-\lambda x} (\lambda x)^{\alpha-1} = \lambda \left(\frac{1}{\lambda} (\lambda x) \right) e^{-\lambda x} (\lambda x)^{\alpha-1} \end{array} \right. \\ &= \frac{\Gamma(\alpha+1)}{\lambda \Gamma(\alpha)} \\ &= \frac{\alpha}{\lambda} \end{aligned}$$

$$\text{Var}(X) = \frac{\alpha}{\lambda^2}$$

The Gamma Distribution is used in real world applications where data is strictly positive, right skewed, and models cumulative wait times.

• The Weibull Distribution

- widely used in engineering, survival analysis, and extreme value theory to model lifetime time-to-time failure. (like consider object with many parts, can experience death when any part fails)

Takes the form

$$F(x) = 1 - e^{-\left(\frac{x}{\lambda}\right)^k}$$

ex A batch of newly manufactured microchips exhibit defect driven early failures, modeled with $k = .5$ (decreasing failure rate) and $\lambda = 1000$ hours.
What % fail within the 100-hour burn in test?

$$F(100) = 1 - e^{-\left(\frac{100}{1000}\right)^{-0.5}} = .2711 \quad , \quad 27.1\% \text{ fail. Chips surviving have a much higher chance going forward.}$$

Note: k is failure rate, λ here is the scale (in Poisson/Exponential it is rate)

• Cauchy Distribution

- no defined $E[X]$ or $\text{Var}(X)$
- $(-\infty, \infty)$

$$f(x) = \frac{1}{\pi} \cdot \frac{1}{1 + (x - \theta)^2} \quad -\infty < x < \infty, \quad \text{parameter } \theta: (-\infty, \infty)$$

$$F(x) = \int f(x) = \frac{1}{\pi(1+x^2)}$$

• Pareto Distribution

- in human productivity, a very tiny percentage of people produce all the output
- ↳ If you look at the # of people who are producing in a given domain, the square root of the people produce half the product
- ↳ money (think Monopoly) - the inevitable consequence where the minority have everything

If X is an exponential r.v. with rate λ and $a > 0$, then $Y = ae^X$ is said to be a Pareto r.v. with parameters a and λ , minimum and index parameter, respectively.

$$P(Y \geq a) = 1$$

$$\begin{aligned} P(Y > y) &= P(ae^X > y) \\ &= P(e^X > y/a) \\ &= P(X > \log(y/a)) \\ &= e^{-\lambda \log(y/a)} \\ &= e^{[-\log(y/a)]^\lambda} \\ &= (a/y)^\lambda \end{aligned}$$

Distribution Function of Y :

$$\begin{aligned} F_Y(y) &= 1 - P(Y > y) \\ &= 1 - a^\lambda y^{-\lambda}, \quad 1 - (a/y)^\lambda \end{aligned}$$

Differentiating yields the density:

$$f_Y(y) = \lambda a^\lambda y^{-(\lambda+1)}, \quad \frac{\lambda a^\lambda}{y^{\lambda+1}}, \quad y \geq a$$

$$\begin{aligned} \text{When } \lambda \leq 1, E[Y] &= \infty \\ \lambda > 1, E[Y] &= \int_a^\infty \lambda a^\lambda y^{-\lambda} dy \\ &= \lambda a^\lambda \frac{y^{1-\lambda}}{1-\lambda} \Big|_a^\infty \\ &= \lambda a^\lambda \frac{a^{1-\lambda}}{\lambda-1} \\ &= \frac{\lambda a}{\lambda-1} \end{aligned}$$

$$\begin{aligned} E[Y^2] &= \int_a^\infty \lambda a^\lambda y^{1-\lambda} dy \\ &= \lambda a^\lambda \frac{y^{2-\lambda}}{2-\lambda} \Big|_a^\infty \\ &= \frac{\lambda a^2}{\lambda-2} \end{aligned}$$

$$\text{When } \lambda > 2, \text{Var}(Y) = \frac{\lambda a^2}{\lambda-2} - \frac{\lambda^2 a^2}{(\lambda-1)^2} = \frac{\lambda a^2}{(\lambda-2)(\lambda-1)^2}$$

Y : actual outcome/data point

↳ income

a : minimum threshold, cannot fall under

λ : shape param/ inequality knob, how heavy is the tail?

ex: Y : individual wealth

$a = 50k$, we are seeing dist of

people with at least 50k

$\lambda = 1$ (small) gives a fat tail
where few pull massive sums

Suppose $a = 10$, $P(Y > 100)$? (10x baseline)

$$P(Y > 100) = \left(\frac{10}{100}\right)^\lambda = (0.1)^\lambda$$

If $\lambda = 1$, $P(Y > 100) = (0.1)^1 = 10\%$

$\lambda = 2$, $P(Y > 100) = (0.1)^2 = 1\%$

$\lambda = 3$, $P(Y > 100) = (0.1)^3 = 0.1\%$

1 in 10 observations a exceed 100

1 in 100 a exceed 100

Adjusting λ shifts the world from being dominated by rare mega events ($\lambda \approx 1$) versus a world where every sample sits right on top of minimum a .

5.7 Distribution of Function of a R.V.

Suppose we have distribution X and want to find the distribution $g(X)$. To do so, it's necessary to express that $g(X) \geq y$ in terms of X

ex: Let X be uniformly distributed over $(0,1)$. We obtain distribution of the r.v. Y defined by $Y = X^n$ as follows: for $0 \leq y \leq 1$

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(X^n \leq y) \\ &= P(X \leq y^{1/n}) \\ &= F_X(y^{1/n}) \\ &= y^{1/n} \end{aligned}$$

⇒ so the density func:

$$f_Y(y) = \begin{cases} \frac{1}{n} y^{1/n-1} & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

ex: If X is a continuous r.v. with pdf f_X , then the distribution $Y = X^2$ is as follows:

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(X^2 \leq y) \\ &= P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= F_X(\sqrt{y}) - F_X(-\sqrt{y}) \end{aligned}$$

Theorem: X is r.v. with pdf f_X . Suppose $g(x)$ is strictly monotonic, differentiable function of x . Then the r.v. Y defined by $Y = g(X)$ has pdf given by

$$f_Y(y) = \begin{cases} f_X[g^{-1}(y)] \left| \frac{d}{dy} g^{-1}(y) \right| & \text{if } y = g(x) \text{ for some } x \\ 0 & \text{otherwise} \end{cases}$$